Proceedings of Machine Learning Research 275:1–27, 2026 5th Conference on Causal Learning and Reasoning

A Computationally Feasible Framework for Causal Probabilistic Explanation

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Editor: Under Review for CLeaR 2026

Abstract

Causal attribution and explanation are mainstays of philosophical, scientific, and policy analysis. Some current frameworks, however, such as actual causality, suffer from an inability to scale causal reasoning to large problems, while attribution methods that do scale—such as SHAP or LIME—are not causally aware. To address this two-fold problem, we present Probabilistic Causal Impact (PCI). Inspired by Halpern and Pearl's actual causality and Pearl's probability of necessary and sufficient causation, PCI provides tractable causal explanations in causal probabilistic machine learning models. By the same token, PCI provides explanations for events in the world, insofar as the model adequately represents real mechanisms. By re-conceptualizing explanation as an estimation problem on an expanded, causal probabilistic model, PCI allows for the use of well-established approximation algorithms, unlocking computationally tractable, truly causal explanations for large models. We test PCI on scaled-up, but canonical test cases for actual causation involving overdetermination and undercutting, finding that we recover results deemed intuitive by the literature, and illustrate that the correct behavior is preserved in the continuous version of the problem. We apply PCI to a real-life causal spatio-temporal model with complex machine-learned components trained on millions of data points. In this setting, we compare PCI to the non-causal (yet scalable) attribution method SHAP, and find that our approach yields explanations that agree with domain expert opinion, while SHAP does not.

Keywords: Probabilistic Causal Models, Explanation, Responsibility, Actual Causality, Probability of Necessity and Sufficiency

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1. Introduction

Causal attribution is a central reasoning task in many important problems, such as gauging the efficacy of medical treatments (Shepherd et al., 2024), pinpointing drivers of economic growth (Cheshire and Magrini, 2009), root cause analysis in automated systems (Papageorgiou et al., 2022), or assessing the impact of public policy (Heckman and Pinto, 2022). In general, we may want to understand why an adverse outcome occurred to prevent it from happening again, or understand how and why a predicted, counterfactual positive outcome can be brought about. However, in the absence of scalable, formal causal attribution tools, analysts often resort to either ad-hoc methods, such as finding strong correlation, historical comparison, domain expertise, or non-causal attribution like Granger causality (Granger, 1969) and SHAP (Lundberg and Lee, 2017). Even researchers with an advanced command of statistical and experimental methods may not possess a thorough understanding of how to ascribe cause (Shepherd et al., 2024), and would thus benefit from scalable automation. Here we present a tool for accurate and tractable causal attribution that can be applied to a wide array of causal, probabilistic, machine-learned models.

Current methods for attributing causes to specific outcomes, such as actual causality (Halpern, 2016), are intractable in general, with no clear approximation (Eiter and Lukasiewicz, 2002). Explanation methods that *do* scale are either non-causal (Lundberg and Lee, 2017) or amount to local, gradient-based sensitivity analyses (Butler et al., 2022). SHAP (Lundberg and Lee, 2017) grounds attribution in observational correlations, so a feature can receive high attribution merely for being correlated with a true cause. Causal SHAP (Heskes et al., 2020) addresses this to some extent, but some failure modes persist (Section 4). Butler et al. (2022)'s Differential Causal Effect (DCE) relies on gradients, which requires model differentiability and does not capture effects from non-infinitesimal changes in potential causes. (DCE is, fairly, not designed for responsibility attribution: Butler et al. target rates of change of treatment.)

In contrast, we introduce probabilistic causal impacts (PCI), which exploit probabilistic understandings of context and causation to frame attribution as a turnkey estimation problem. In particular, we generalize Pearl's formulation of the "probability of necessary and sufficient causation" (Pearl, 2022) for arbitrary variable domains, and additionally incorporate causal context in the form of "witnesses" as understood by actual causality researchers (Halpern, 2016). Our construction makes partial contact with the study of *descriptive normality* (Halpern and Hitchcock, 2015; Icard et al., 2017), which holds that causal attributions depend on a graded measure of which counterfactual scenarios are plausible. We characterize attribution as the expectation of a customizable causal impact function with respect to a distribution over counterfactual worlds, where this distribution is conditioned on upstream factual context — a concrete, if partial, operationalization of the normality intuition. PCI combines a variety of intuitions from the literature, including taking expectations over exogenous inputs (cf. Halpern (2016) on "blame", def. 6.2.1; and implicitly by Pearl's (2022) probabilities of causation), weighting by the probability of alternative values for the supposed cause (Beckers and Vennekens, 2015, def. 8), aggregating the probabilities of all valid witness worlds (Beckers and Vennekens, 2015, §6.2), or weighting by the complexity of the least complex witness set (Halpern, 2016, on "responsibility", def. 6.1.6). Unfortunately, taken individually, none of these existing approaches achieves the scalability and generality required for modern causal attribution needs, thereby necessitating the development of PCI.

PCI presumes access to a trained probabilistic causal model from which counterfactual outcomes can be sampled — a stronger requirement than methods that work from observational data and a causal graph alone (e.g., PN/PS/PNS bounds, Causal SHAP). Within that requirement, however, we place no strong restrictions on the form of the model, distributions, or types of variables. Such models can comprise standard machine learning components such as neural networks or Gaussian processes.

With this breadth of application, and a measure that surfaces its design choices — the variable selection distribution, alternative value distribution, and causal impact function — as explicit user-facing components rather than hidden defaults, PCI hopes to provide causal attribution and explanation tooling for a wide array of users and applications.

To validate and better understand the approach, and to illustrate how our method scales to far larger models than does exact actual causality computation, we take the following steps:

revise this list once paper done

1. We draw theoretical connections to actual causality in Section **??**, proving that under certain constraints the diagnoses of actual causality and PCI coincide.
2. In Section **??**, we test PCI on scaled-up, but canonical failure cases for actual causation involving overdetermination and undercutting, finding that we recover results deemed intuitive by the literature,
3. In Section **??** we point also to the correct behavior in the continuous version of the problem.
4. We also delegate a more extensive conceptual discussion of the relation between PCI , actual causality, and probabilities of necessity, sufficiency and necessity and sufficiency to Appendix **??**.
5. In Section **??** we describe results on a causal spatio-temporal model with complex machine-learned components trained on millions of data points. In this setting, we compare to the non-causal but scalable attribution method SHAP and discuss the differences in the resulting explanations.
6. Additionally, in Appendix **??**, we compare the local, sensitivity-analysis-style attribution from Butler et al. (2022), demonstrating cases where PCI better reveals causality.

2. Causal Impact: motivations

The central difficulty in causal attribution is not computing counterfactuals — it is knowing which counterfactuals to compute, and recognising when a cause is blocked by context rather than absent altogether. The closest predecessors of this approach, taken individually, fail to capture this: actual causality handles context sensitivity but is computationally intractable and not general enough; the probability of necessity and sufficiency scales well, but lacks context sensitivity. This section builds intuition for both limitations through a running example, setting up the design choices that motivate PCI in Section 3.

2.1. Actual Causality

Consider the following (deterministic, discrete) model:

Example 1 (Bob at Old Boys' Club Bank) *Bob applied for a home mortgage loan with the local branch of Old Boys' Club Bank (OBCB), and his application was denied. The OBCB policy requires applicants to pass a credit score check, but it only performs this check if the applicant is male; all* *non-male applicants are rejected outright. Thus, the procedure is:*

$$check = (gender = male)$$

$$check\text{-}failed = check \wedge (credit = bad)$$

$$loan = \neg check\text{-}failed \wedge (gender = male)$$

Why was Bob's loan application denied? An intuitive explanation is that his credit was bad—had it been good, the outcome would have been different. In contrast, his gender does not appear to be responsible for the rejection: had it been different, his application still would have been denied (albeit this time due to $check = F$).

This suggests a counterfactual approach to feature responsibility attribution: perhaps we can identify the features responsible for a particular outcome by asking counterfactual questions—would the outcome have been different if a specific feature had a different value? This is called the *but-for clause*: Bob wouldn't have been denied the loan but for his bad credit.

Concretely, consider a model M with *exogenous variables* $\mathbf{U}$, which are the source of stochasticity, and *endogenous variables* $\mathbf{V}$, which are influenced by other variables in the model. Let $\mathbf{S} \subseteq \mathbf{V}$ be the variables we suspect to be causally responsible (call them *causal suspects*), and $Y \in \mathbf{V}$ be an outcome of interest. Suppose we consider only one suspect $S \in \mathbf{S}$ and mark factual values with *. Under a setting $\mathbf{U} = \mathbf{u}$ of the exogenous variables, M produces $S = s^\star$ and $Y = y^\star$, denoted $(M, \mathbf{u}) \models S = s^\star, Y = y^\star$. We call this the *factual* scenario. An intervened model where a variable S is set to an alternative value s' is denoted $(M, \mathbf{u})[S \leftarrow s']$.

A first attempt at defining a counterfactual notion of causal responsibility is:

Definition 1 (But-for, single) *$S = s^\star$ is causally responsible for $Y = y^\star$ in $(M, \mathbf{u})$ if and only if $(M, \mathbf{u}) \models S = s^\star, Y = y^\star$ and $(M, \mathbf{u})[S \leftarrow s'] \not\models Y = y^\star$ for some $s' \neq s^\star$.*

In a given context, S is causally responsible for an outcome only when intervening on S with some alternative value would change that outcome. Bob's credit score satisfies this condition: had his credit score been higher, the loan might have been approved. However, this cause test is too coarse.

Consider another case:

Example 2 (Alice at OBCB) *Alice is a woman with bad credit. She applies for a loan with OBCB and is rejected.*

Why was Alice denied the loan? Under the but-for clause for single cause candidates, neither her gender nor her credit score are responsible. No matter how high her credit score, her application would still have been denied because the bank does not loan to women. Had her gender been different, she would still have been denied for having bad credit.

To address this example, we could consider sets of factors as causes:

Attempt 1 (But-for, sets) *Variables $\mathbf{C} \subseteq \mathbf{S}$ with factual values $\mathbf{C} = \mathbf{c}^\star$ are causally responsible for $Y = y^\star$ in $(M, \mathbf{u})$ if and only if $(M, \mathbf{u}) \models \mathbf{C} = \mathbf{c}^\star, Y = y^\star$ and $(M, \mathbf{u})[\mathbf{C} \leftarrow \mathbf{c}'] \not\models Y = y^\star$ for some $\mathbf{c}'$ everywhere different from $\mathbf{c}^\star$, where no proper subset of $\mathbf{C}$ has this property.*

The minimality condition reflects our preference for explanations that include only relevant properties.

Under this generalization, the set $\mathbf{c}^\star = \{gender = \text{female}, credit = \text{bad}\}$ is jointly responsible for Alice's application being denied because there exists $\mathbf{c}' = \{gender = \text{male}, credit = \text{good}\}$ that would change the decision and $\mathbf{c}^\star$ is minimal. But what can we say about the individual features? We could say that the whole set bears responsibility but individual features do not, or that any feature in a cause set is partially and equally accountable. Both approaches have issues, as they imply the features are equally responsible. This conflicts with our intuition: OBCB did not evaluate Alice's credit score, so her denial was caused by her gender.[1]

Prior work suggests a way out of this difficulty: hold some of the context in which counterfactuals are evaluated fixed, calling such fixed variables *witnesses*. Let us focus on the mediator $check\text{-}failed$ between $gender$ and $loan$. If we fix this variable at the factual value by intervening, then counterfactually, Alice would have been granted the loan if she were male. However, there is no context fixed at the actual value such that, had her credit score been higher, she would have been granted the loan. This motivates the following formalization:

Definition 2 (Actual cause) *Let $\mathbf{C} \subseteq \mathbf{S}$ be a cause set, $\mathbf{W} \subseteq \mathbf{V}$ be a set of possible witnesses. $\mathbf{C} = \mathbf{c}^\star$ is causally responsible for $Y = y^\star$ in $(M, \mathbf{u})$ just in case there is a $\mathbf{T}$ (an actual witness set to be held fixed) s.t. $\mathbf{T} \subseteq \mathbf{W}, \mathbf{T} \cap \mathbf{C} = \emptyset$ and $(M, \mathbf{u}) \models \mathbf{C} = \mathbf{c}^\star, Y = y^\star$ and $(M, \mathbf{u})[\mathbf{C} \leftarrow \mathbf{c}', \mathbf{T} \leftarrow \mathbf{t}^\star] \not\models Y = y^\star$, for some $\mathbf{c}'$ distinct everywhere from $\mathbf{c}^\star$, where no proper subset of $\mathbf{C}$ satisfies this property.*

This definition extends but-for causality with a witness set $\mathbf{T}$ that is fixed to its factual value by an intervention. Under this definition, to determine the cause of an outcome, we need to find a cause set $\mathbf{C}$, alternative values $\mathbf{c}'$ (trivial if the variables are binary and less so if they are categorical or continuous), and a witness set $\mathbf{T}$. $\mathbf{C}$ and $\mathbf{T}$ are drawn from sets of size at most $2^{|\mathbf{V}|}$, so a brute-force computation of these sets is expensive.

Putting computational difficulties aside, we have so far considered deterministic models, but would like to capture probabilistic models as well, so we now consider a stochastic variant of the OBCB problem.

2.2. Probability of Necessity and Sufficiency

Example 3 (OBCB, stochastic) *The bank performs credit checks for a randomly selected 20% of women and skips checks for a randomly selected 10% of men. Unchecked applicants are rejected. Furthermore, if a man's credit check fails, he is nevertheless accepted 5% of the time and if a woman's credit check succeeds, she is denied 10% of the time. We assume that the population is evenly divided into male and female and good and bad credit. Formally, the model, using 1 for male and 1 for good credit, is:*

1. Examples of this sort, involving both over-determination and undercutting of causal impact, abound. Suppose Jim is both stabbed and poisoned and dies as a result. Because stabbing works quickly, it kills Jim before the poison takes effect. In this case, it's the stabbing that caused his death. Yet analogous but-for clauses fail to break the symmetry here as well.

$$
\textit{loan-prob} = \begin{array}{r|cc}
 & \textit{check-failed}{=}0 & \textit{check-failed}{=}1 \\
\hline
\textit{gender}{=}0 & 0.9 & 0 \\
\textit{gender}{=}1 & 1 & 0.05
\end{array}
$$

$$
\begin{aligned}
\textit{credit} &\sim \mathrm{Bern}(0.5) \\
\textit{gender} &\sim \mathrm{Bern}(0.5) \\
\textit{check} &\sim \mathrm{Bern}((1-\textit{gender})\cdot 0.2 + \textit{gender}\cdot 0.9) \\
\textit{check-failed} &= \textit{check}\cdot(1-\textit{credit}) \\
\textit{loan-if-checked} &\sim \mathrm{Bern}(\textit{loan-prob}[\textit{gender}, \textit{check-failed}]) \\
\textit{loan} &= \textit{loan-if-checked}\cdot\textit{check}
\end{aligned}
$$

The witness mechanism works well for deterministic models, but actual causality in its standard form offers no straightforward way to handle probabilistic ones. Let's shelve that problem for now and instead ask: what would a probabilistic approach to causal attribution look like, and what would it miss without witnesses?

Before extending the witness idea to the stochastic setting, it is instructive to examine a probabilistic approach that scales more naturally: Pearl's probabilities of necessity and sufficiency (Pearl, 2022). This will let us identify what probabilistic machinery is needed, and what context-sensitivity is lost when witnesses are absent — motivating the synthesis that PCI provides. The following definitions allow for probabilistic models, but assume that all variables are Boolean (PCI will relax this assumption, allowing for continuous variables). Pearl considers two ways that a cause may relate to an effect: it may be necessary or sufficient for the effect. A cause is necessary if, counterfactually absent the cause, the effect is unlikely to occur; it is sufficient if the effect is counterfactually likely in the presence of the cause. We now formalize these ideas and discuss them in the context of the running example.

Definition 3 (Probability of Necessity) *Let the probability of necessity be the probability that the outcome $Y = y$ would not have occurred in the absence of $C = c^\star$, given that both occurred:*

$$PN(C = c^\star, Y = y^\star) = P(Y_{c'} \neq y^\star \mid C = c^\star, Y = y^\star)\,,$$

where $Y_{c'}$ is the value of Y after intervening on C with $c' \neq c^\star$.

Strictly speaking, the probability of necessity is less specific to a particular situation than the actual cause test discussed above. For example, we can consider the probability that being male causes someone to fail to get a loan, but we can't consider the probability that being male causes Bob specifically (who we know has bad credit) to fail to get a loan. We might try to patch this up by conditioning, for the time being.

We illustrate the computation for Alice's gender. Alice is factually $\textit{gender}$ = female, $\textit{credit}$ = bad, $\textit{loan} = F$; the counterfactual intervenes to set $\textit{gender}$ = male while leaving $\textit{credit}$ = bad. Tracing through the model:

- Under the intervention, $\textit{check} \sim \mathrm{Bern}(0.9)$ — men are checked 90% of the time.

- If checked, $\textit{check-failed} = 1$ (Alice's credit is bad), and from the loan-probability table $P(\textit{loan-if-checked} = 1 \mid \textit{gender} = \text{male}, \textit{check-failed} = 1) = 0.05$.

- If not checked, $\textit{loan} = 0$ deterministically since $\textit{loan} = \textit{loan-if-checked} \cdot \textit{check}$.

Combining the two branches:

$$
\begin{aligned}
&PN(\textit{gender} = \text{female}, \textit{loan} = F \mid \text{Alice}) \\
&\qquad = P(\textit{loan}_{\textit{gender}=\text{male}} = T \mid \textit{gender} = \text{female}, \textit{credit} = \text{bad}) \\
&\qquad\qquad = 0.9 \cdot 0.05 = 0.045.
\end{aligned}
$$

Computing the analogous quantities for the remaining (variable, person) pairs gives the individual-level values shown in Table 1; the table also lists population-level values for comparison.

The PN scores partially align with intuition: Bob's bad credit is high ($PN = 0.90$), and his gender score is zero, matching the reading that his gender played no role. But Alice's PN values do not separate gender (0.045) from credit (0.18) particularly well, and Alice's gender PN is only marginally above Bob's — not the sharp distinction we would want, given that the bank never even evaluated Alice's credit. PN cannot register this asymmetry: it asks counterfactually what would happen under intervention without any context about which causal paths were active. For the full picture we need a second quantity, the probability of sufficiency.

Definition 4 (Probability of Sufficiency) *Let the probability of sufficiency be the probability that the outcome $Y = y^\star$ would have occurred in the presence of $C = c^\star$, given that neither $C = c^\star$ nor $Y = y^\star$ occurred:*

$$PS(C = c^\star, Y = y^\star) = P(Y_{c^\star} = y^\star \mid C = c',\ Y \neq y^\star)$$

where $c' \neq c^\star$ and $Y_{c^\star}$ is the value of Y after intervening on C with $c^\star$.

To query whether the factual cause was sufficient, we condition on the counterfactual: we assume the cause did *not* take its factual value and the outcome did *not* occur, then intervene to restore the factual cause value and ask whether the outcome now occurs. Intuitively, this asks: if the cause had been absent and things had gone differently, would reinstating the cause have brought the outcome about? Let's look at the resulting values.

We illustrate PS for Alice's gender. Following the same convention as for PN, the individual-level computation additionally conditions on Alice's credit ($\textit{credit} = \text{bad}$). The PS conditional then has Alice's credit fixed but the cause and outcome flipped to their non-factual values ($\textit{gender} = \text{male}$, $\textit{loan} = T$); we intervene to restore $\textit{gender} = \text{female}$ and ask whether the loan is again denied:

- With $\textit{gender} = \text{female}$ and $\textit{credit} = \text{bad}$, $\textit{check} \sim \mathrm{Bern}(0.2)$.

- If checked, $\textit{check-failed} = 1$, and from the loan-probability table $P(\textit{loan-if-checked} = 1 \mid \textit{gender} = \text{female}, \textit{check-failed} = 1) = 0$.

- If not checked, $\textit{loan} = 0$ deterministically.

Either branch gives $loan = F$, so

$$\begin{aligned}&PS(gender = \text{female}, loan = F \mid \text{Alice})\\ &\qquad\qquad\qquad = P(loan_{gender=\text{female}} = F \mid gender = \text{male}, loan = T, credit = \text{bad}) = 1.\end{aligned}$$

The remaining individual values, along with population-level values for comparison, appear in Table 1. Two cases warrant comment: Bob's gender PS is undefined because the required conditioning event ($gender$ = female, $credit$ = bad, $loan = T$) has probability zero in the model; Bob's credit PS is 0.955 rather than 1 because the model includes a 5% exception rate for males who fail the credit check.

To somehow combine insights provided by the two scores, we would like to find causes that are both necessary and sufficient. This notion is formalized as follows:

Definition 5 (Probability of necessity and sufficiency) *Let the probability of necessity and sufficiency be the probability that the outcome $Y = y^\star$ would have occurred in the presence of $C = c^\star$ and that it would not have occurred under the intervention $C = c'$:*

$$PNS(C = c^\star, Y = y^\star) = P(Y_{c^\star} = y^\star, Y_{c'} \neq y^\star)$$

The population- and individual-level PNS values appear in the rightmost column group of Table 1. These values capture the relative causal impact of $gender$ and $credit$ but are not entirely satisfying. Table 1 reveals the problem directly: for Alice, $PNS(gender) = 0.045 < PNS(credit) = 0.18$, so PNS ranks credit as more causally responsible than gender. Yet the bank never evaluated Alice's credit—the $check$ variable was F throughout. PNS has no mechanism to register this: it asks counterfactually what would happen if credit were different, without fixing $check\text{-}failed$, the mediating variable that blocked credit's causal path.

As an attempt to help Alice better understand her particular situation, we might consider a modification of PNS with additional conditioning:

$$PNS_c(C = c^\star, Y = y^\star \mid F = f^\star) = P(Y_{c^\star} = y^\star, Y_{c'} \neq y^\star \mid F = f^\star)$$

Using this definition, we can compute $PNS_c(gender = \text{female}, loan = F \mid credit = \text{bad}) = 0.045$ and $PNS_c(credit = \text{bad}, loan = F \mid gender = \text{female}) = 0.18$. Conditioning does not solve the problem described above—Alice's credit was in reality never evaluated—because the interventions override the mediating variable that tracks whether her credit has been checked. When we compute $PNS_c(credit = \text{bad}, loan = F \mid gender = \text{female})$, conditioning on gender being female is consistent with the factual—but when we then intervene to set credit to good, this intervention propagates downstream and changes $check\text{-}failed$, overriding what we learned from conditioning on Alice's factual outcome. The context we fixed is undone by the very intervention meant to probe it. We will not consider this use of conditioning further, and this observation motivates the need for intervention-based (as opposed to conditioning-based) context-sensitivity, which is what the witness mechanism provides.

2.3. Path forward

What should we conclude from this investigation of actual causality and the probability of necessity and sufficiency? First, we need a way to treat probabilistic models, and the probability of necessity

Table 1: PN, PS, and PNS for Alice and Bob in Example 3, with $loan = F$ as the outcome. "Population" values condition only on the cause–outcome pair prescribed by the definition; "Individual" values additionally condition on the subject's other factual variable (e.g. for Alice's gender, on $credit$ = bad). A dash marks an entry that is undefined because the conditioning premises are inconsistent with the model.

		Alice			Bob		
Variable	Level	PN	PS	PNS	PN	PS	PNS
$gender$	Population	0.43	0.83	0.39	0.02	0.10	0.01
$gender$	Individual	0.045	1.00	0.045	0	—	0
$credit$	Population	0.53	0.96	0.52	0.53	0.96	0.52
$credit$	Individual	0.18	1.00	0.18	0.90	0.955	0.86

and sufficiency gives us a tool for doing that. However, PNS is not as specific as we would like because it does not capture relevant features of individual units that are not part of the cause. Actual causality offers more specificity, but requires modification to deal with probability and continuous variables, and is computationally expensive to compute.

PCI addresses both limitations by combining the two approaches. To see the intuition, consider Alice's case once more. When PCI evaluates gender's causal impact on her denial, it considers counterfactual worlds in which her gender is changed, while drawing witness sets from a distribution that includes $check\text{-}failed$ at its factual value of F — the credit check was never performed. In those worlds, changing gender to male, with $check\text{-}failed$ held fixed at F, leads to loan approval: a large counterfactual shift, and so high attribution. When PCI evaluates credit's causal impact, it searches for witness sets that, held fixed, would make a change to good credit matter for Alice — but no such context exists, because the credit check was never run. The distribution over witness sets therefore assigns low weight to any world in which credit is relevant for Alice, and her credit receives low attribution. Unlike actual causality, PCI does not ask whether *some* witness set validates a counterfactual dependency; it integrates over the distribution of all witness sets, weighting each by its probability. Unlike PNS, it inherits the context-sensitivity of actual causality through this integration. The formal definitions in Section 3 make this expectation precise — but the intuition is the one we have just seen: PCI inherits the witness mechanism of actual causality and the probabilistic language of PN/PNS, and combines them into a tractable estimation problem.

3. Causal Impact: definitions

Explanation and causal attribution seek to connect a potentially causal *event* $X = x$ to an outcome event $Y = y$.[2] In this section, we present a formal development. To ground it, we return to Example 3 and use Alice and Bob as running illustrations throughout this section. Both are rejected for a loan; the question is which of their attributes—$gender$ and $credit$—was causally responsible for each rejection, and to what degree. While the example is discrete, the machinery applies to continuous and mixed-type variables; a continuous illustration is given in Section **??**.

Add cross-reference to continuous example.

2. This is quite different from type-level causal queries, where one is interested in some group-level attribution, as when one estimates average treatment effect or conditional average treatment effect.

Let $R(V \rightsquigarrow \mathit{loan} \mid \text{person})$ denote the causal responsibility of variable V for the loan outcome at the individual level (R is a placeholder for any attribution score and should not be identified with a specific formal quantity; PCI interprets R as $\mathbb{E}[ci(Y^s, Y^n, y^\star)]$, but the desiderata are method-agnostic). Attribution methods may produce signed values; the desiderata are stated in terms of absolute magnitudes $|R(\cdot)|$, allowing for directionality while remaining agnostic about sign conventions. The following desiderata encode what an intuitively correct attribution should look like:

D-A1 $|R(\mathit{gender} \rightsquigarrow \mathit{loan} \mid \text{Alice})| > 0$

D-A2 $|R(\mathit{credit} \rightsquigarrow \mathit{loan} \mid \text{Alice})| > 0$

D-A-rank $|R(\mathit{gender} \rightsquigarrow \mathit{loan} \mid \text{Alice})| > |R(\mathit{credit} \rightsquigarrow \mathit{loan} \mid \text{Alice})|$

D-B1 $|R(\mathit{gender} \rightsquigarrow \mathit{loan} \mid \text{Bob})| > 0$

D-B2 $|R(\mathit{credit} \rightsquigarrow \mathit{loan} \mid \text{Bob})| > 0$

D-B-rank $|R(\mathit{credit} \rightsquigarrow \mathit{loan} \mid \text{Bob})| > |R(\mathit{gender} \rightsquigarrow \mathit{loan} \mid \text{Bob})|$

D-comp $|R(\mathit{gender} \rightsquigarrow \mathit{loan} \mid \text{Alice})| > |R(\mathit{gender} \rightsquigarrow \mathit{loan} \mid \text{Bob})|$

D-A1 and **D-A2** both require non-zero attribution for Alice because two distinct causal routes were available: gender filtered her out at the checking stage without her credit ever being evaluated, while bad credit would have mattered had she been checked. Neither is causally inert. **D-A-rank** is the substantive constraint: because Alice never underwent a credit evaluation, gender's causal role is immediate and structural, whereas credit's is purely hypothetical. A method that ranks credit above gender for Alice is misattributing the cause of her rejection.

For Bob, **D-B2** reflects that the credit check occurred and directly produced the rejection: the causal chain $\mathit{credit} \rightarrow \mathit{check\text{-}failed} \rightarrow \mathit{loan}$ was activated. **D-B1** requires strictly positive attribution for gender: being male gave Bob a 90% probability of being checked, so gender opened the evaluation that rejected him. This indirect causal role is non-zero; a method returning exactly zero fails to capture it. **D-B-rank** corresponds to the intuition that credit is the proximate cause. **D-comp** captures the cross-individual asymmetry that is hardest for naive methods to express: gender is more causally responsible for Alice's rejection than for Bob's, because for Alice gender=female was sufficient on its own to produce the rejection — no evaluation took place, so credit was never a factor — while for Bob gender=male merely opened the evaluation that credit=bad then caused. We will verify at the end of this section that PCI satisfies all seven desiderata, while PNS fails **D-A-rank** and **D-B1**.

We work within the context of probabilistic structural causal models (Pearl, 2009).

Definition 6 (Probabilistic Structural Causal Model (PSCM)) *A probabilistic structural causal model is a tuple*

$$M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle,$$

where $\mathbf{V} = \{V_1, \ldots, V_n\}$ *is a finite set of endogenous variables,* $\mathbf{U} = \{U_1, \ldots, U_n\}$ *is a set of exogenous variables, and* $\mathbf{F} = \{f_1, \ldots, f_n\}$ *is a set of structural assignments*

$$V_i := f_i(\mathbf{Pa}_i, U_i), \qquad i = 1, \ldots, n,$$

with $\mathbf{Pa}_i \subseteq \mathbf{V} \setminus \{V_i\}$ *denoting the parents of* V_i*. The distribution* $P_{\mathbf{U}}$ *is a probability measure on* $\mathrm{dom}(\mathbf{U})$*, and the assignments in* $\mathbf{F}$ *are assumed to be acyclic and to have a unique solution for* $\mathbf{V}$ *given* $\mathbf{U}$*.*[3]

A PSCM encodes which variables influence which others, through what mechanisms, and subject to what randomness. The structural equations specify each variable as a deterministic function of its causal parents and local noise; the exogenous distribution captures all irreducible uncertainty in the system. Together they make every counterfactual question well-posed and computable.

The OBCB model (Example 3) is a PSCM with endogenous variables

$$\mathbf{V} = \{\mathit{gender}, \mathit{credit}, \mathit{check}, \mathit{check\text{-}failed}, \mathit{loan}\},$$

and exogenous variables

$$\mathbf{U} = \{U_{\mathit{gender}}, U_{\mathit{credit}}, U_{\mathit{check}}, U_{\mathit{loan}}\}$$

which induce independent Bernoulli noise.

The structural assignments in the case at hand can be written as:

$$\begin{aligned} \mathit{gender} &\sim \mathrm{Bernoulli}(0.5), \qquad \mathit{credit} \sim \mathrm{Bernoulli}(0.5), \\ \mathit{check} &\sim \mathrm{Bernoulli}\big(0.2(1-\mathit{gender}) + 0.9\mathit{gender}\big), \\ \mathit{check\text{-}failed} &:= \mathit{check}(1-\mathit{credit}), \\ \mathit{loan} &\sim \mathrm{Bernoulli}\big(\text{loan-prob}[\mathit{gender}, \mathit{check\text{-}failed}]\big) \cdot \mathit{check}. \end{aligned}$$

Definition 7 (Interventions) *Given a model* $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$*, an intervention on variables* $\mathbf{X} = \{X_1, \ldots, X_k\} \subseteq \mathbf{V}$ *with values* $\mathbf{x} = (x_1, \ldots, x_k)$ *replaces the structural assignments for* X_i *by* $X_i := x_i$*, where* $i = 1, \ldots, k$*. We denote this intervention by either* $[\mathbf{X} \leftarrow \mathbf{x}]$ *or* $do(\mathbf{X} = \mathbf{x})$*.*

An intervention surgically replaces a variable's structural equation with a fixed constant, disconnecting it from its usual causes. The rest of the model propagates downstream as normal under the new assignment; the exogenous noise is unchanged.

In the OBCB model, $\mathrm{do}(\mathit{gender} = 1)$ replaces Alice's gender assignment with male, leaving all other structural equations intact. The downstream effect propagates: check now draws from $\mathrm{Bern}(0.9)$ rather than $\mathrm{Bern}(0.2)$, so Alice would be checked in 90% of noise realizations instead of 20%. This is the counterfactual world in which we ask whether her loan outcome would have differed.

3. Working within the PSCM framework commits the practitioner to fully specified structural assignments $\mathbf{F}$ and an exogenous distribution $P_{\mathbf{U}}$. This is a standard feature of probabilistic model-based causal inference — analogous to how Bayesian inference requires a likelihood and a prior — and it is precisely what enables PCI to yield individual-level counterfactual attributions rather than population-level bounds. Sampling from $P_{\mathbf{U}}$ is then a routine probabilistic inference operation. This requirement is a scope limitation worth stating explicitly. Standard Shapley values (Lundberg and Lee, 2017) require only expectations of the form $\mathbb{E}[f(\mathbf{X}_S, \mathbf{X}_{\bar{S}})]$ — no causal structure at all. Causal Shapley methods (Heskes et al., 2020) additionally require a causal graph and observational or interventional data, but not the structural equations themselves. Pearl's PN/PS/PNS can be bounded from observational or interventional data given the graph, again without the full noise model. PCI requires strictly more: the complete structural assignments $\mathbf{F}$ and the noise distribution $P_{\mathbf{U}}$. The payoff is individual-level counterfactual attribution — not population-level bounds or marginal contributions averaged over a reference distribution — but practitioners without a fully specified structural model will need to build or assume one.

Because PCI supports arbitrary variable types, it will be helpful to define a measure theoretic object corresponding to interventional outcomes. Because outcomes are a deterministic function of exogenous noise, this will be a Dirac measure.

Definition 8 (Interventional Law) *Let $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$ be a probabilistic structural causal model and let $Y \in \mathbf{V}$. For an intervention $\mathrm{do}(\mathbf{X} = \mathbf{x})$, let $Y' : \mathrm{dom}\,(\mathbf{U}) \to \mathrm{dom}\,(Y)$ denote the measurable map induced by the modified structural assignments. The interventional law of Y given fixed $\mathbf{u} \in \mathrm{dom}\,(\mathbf{U})$ is the probability measure on $\mathrm{dom}\,(Y)$ defined by*

$$P(Y \in A \mid \mathbf{u}, \mathrm{do}(\mathbf{X} = \mathbf{x})) \;:=\; \mathbb{I}\big\{Y'(\mathbf{u}) \in A\big\} \;=\; \delta_{Y'(\mathbf{u})}(A), \qquad A \subseteq \mathrm{dom}\,(Y).$$

The interventional law records the deterministic outcome of Y under $\mathrm{do}(\mathbf{X} = \mathbf{x})$ for a fixed noise realization $\mathbf{u}$: since the modified structural equations make Y a function of $\mathbf{u}$ alone, its distribution collapses to a point mass. Integrating over $P_{\mathbf{U}}$ then recovers the full interventional distribution.

We define a *causal set* to be a set of assignments to the variables of a probabilistic structural causal model.

Definition 9 (Causal set) *Given a model $\langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$, a causal set is a set of pairs $X = x$ where $X \in \mathbf{V}$ and $x \in dom(X)$.*

Informally, a causal set is a list of specific variable-value assignments — a precise statement of which variables are set to which values. It is the basic unit of intervention: from a causal set one can read off exactly what is being manipulated or held fixed.

In PCI , causal sets play four distinct roles. At times, they will represent sets of *potential* causes—whose subsets may be intervened upon— in which case we call them *suspects*, denoted $\mathbf{S}$. When referring to the specific subset we *in fact* intervene on in a given search step, we use $\mathbf{C} \subseteq \mathbf{S}$ and call them *active suspects*. Analogously, some variables serve as *potential* context variables to be held fixed at their actual values; we call these *witnesses*, denoted $\mathbf{W}$. The subset of witnesses actually held fixed in a given step is denoted $\mathbf{T} \subseteq \mathbf{W}$ and called *active witnesses*. When a variable appears in both $\mathbf{C}$ and $\mathbf{W}$, suspect activation takes precedence: $\mathbf{T}$ is pruned so that $\mathbf{T} \cap \mathbf{C} = \emptyset$.

To evaluate the causal impact of a variable on an outcome, we consider two interventional regimes: the *necessity regime*, in which we intervene on the variables $\mathbf{C} \subseteq \mathbf{S}$ to have an alternative value, and the sufficiency regime, in which we intervene so that $\mathbf{C}$ takes on factual values. Moreover, we hold some elements $\mathbf{T} \subseteq \mathbf{W}$ of the context fixed (i.e., intervened on to retain factual values, despite intervention on $\mathbf{C}$). When looking for an explanation, we perform a search across possible cause sets and possible contexts. Complete model samples in these regimes will be called *necessity worlds* and *sufficiency worlds*, respectively.

The intuition is that the suspect set $\mathbf{S}$ names every variable that could plausibly have caused the outcome; the witness set $\mathbf{W}$ names context variables to be selectively held fixed, isolating the causal channels under investigation. Intervening on suspects tests what would have happened under an alternative cause in the necessity worlds and under the factual values in the sufficiency worlds; fixing witnesses blocks confounding paths that would otherwise absorb the effect. Responsibility attribution increases when the outcome is far from the factual value in the necessity world and decreases when it is far from the factual value in the sufficiency world.

For both Alice (female, bad credit, loan denied) and Bob (male, bad credit, loan denied) we set $\mathbf{S} = \{gender, credit\}$ and $\mathbf{W} = \{check\text{-}failed\}$. The witness *check-failed* records whether the applicant was checked and had bad credit; holding it at its factual value isolates the causal path from $gender$ or $credit$ to $loan$ independently of the credit-check bottleneck. Without this witness, intervening on $credit$ for Alice would leave 80% of noise realizations unaffected — those in which she is never checked — making credit's causal role invisible. The witness makes the bottleneck explicit and manipulable.

Moving on with the general framework, we will be searching for causes by intervening on subsets of suspects and holding fixed subsets of witnesses. To make this search tractable, we need a way to prioritize which subsets to consider.

Definition 10 (Variable Selection Distribution) *Let $\mathbf{X}$ be a finite set of random variables. A* variable selection distribution Γ *is a probability distribution on $2^{\mathbf{X}}$. We write $\mathbf{Y} \sim \Gamma$ to mean $\mathbf{Y}$ is a random subset of $\mathbf{X}$ drawn from Γ, with probability mass function $p^{\Gamma}(\mathbf{y}) = \mathbb{P}(\mathbf{Y} = \mathbf{y})$ for $\mathbf{y} \subseteq \mathbf{X}$.*

The variable selection distribution Γ encodes prior beliefs about which subsets of suspects are worth examining. A uniform distribution treats all suspect subsets as equally plausible; a cardinality-constrained choice focuses the search on interactions of a particular order, analogous to choosing which interaction terms to include in a regression. More complex distributions could encode domain knowledge about which variables are more likely to interact, or could be learned from data. The choice of Γ is a design decision that shapes the search process and the resulting attributions. As a default, a uniform Γ over the full powerset is the least-informative starting point, assigning equal weight to all subset sizes, analogous to including all interaction orders in a regression. There is no universally principled criterion for restricting the cardinality range; the honest advice is to treat J and K as explicit design choices. Two practical considerations bear on this choice: with $K = 1$, attribution scores are based on singleton interventions and straightforward to communicate to stakeholders; higher K averages over joint interventions on X_k with other suspects simultaneously, which is harder to explain and, in large or structurally complex models, may amplify estimation noise rather than capturing genuine higher-order causal structure. For applications where attribution has serious consequences, we additionally recommend a cardinality sensitivity analysis: if rankings are stable as K increases from 1 to $|\mathbf{S}|$, the simpler choice is supported; if they diverge, higher-order interactions are materially relevant and the full powerset should be used. The key advantage of PCI is that this commitment is made transparently rather than absorbed into an implicit default.
In particular, we will be using variable selection distributions for the powerset of causal suspects $\mathbf{S}$ and for the powerset of witnesses $\mathbf{W}$. One simple pattern of suspect and witness selection can be derived by setting size limits on the number of activated suspects or witnesses, and sampling uniformly otherwise.

Example 4 (Cardinality-Constrained Uniform Selection) *Let $\mathbf{X}$ be a candidate set of random variables and let $k \sim \mathrm{Unif}\{J, \dots K\}$ with $1 \leq J \leq K \leq |\mathbf{X}|$. A variable selection $\mathbf{Y} \sim \Gamma(2^{\mathbf{X}})$ is sampled from a cardinality-constrained uniform selection distribution $\Gamma(2^{\mathbf{X}})$ with probability mass function*

$$p^{\Gamma}(\mathbf{Y}) \propto \mathbb{I}[|\mathbf{Y}| = k].$$

The parameters J and K determine the cardinality range of the search; specific choices connect PCI to existing methods and enable efficient Monte Carlo approximation.[4]

4. When $J = 1$ and $K = |\mathbf{X}|$, this distribution coincides with the Shapley weighting kernel, which samples coalition size uniformly and then selects a coalition uniformly at that size (Shapley, 1953). The parameters J and K allow the

In PCI , the variable selection distribution Γ determines a pair (Γ_s, Γ_w): Γ_s over $2^{\mathbf{S}}$ governs suspect activation and Γ_w over $2^{\mathbf{W}}$ governs witness selection. The joint necessity-sufficiency measure marginalises over all pairs $(\mathbf{C}, \mathbf{T})$ weighted by $p_s^\Gamma(\mathbf{C})\, p_w^\Gamma(\mathbf{T})$, restricted to $\mathbf{T} \cap \mathbf{C} = \emptyset$.

For the OBCB analysis, with $|\mathbf{S}| = 2$, we use a uniform Γ_s over the three non-empty subsets of $\mathbf{S}$: $\{gender\}$, $\{credit\}$, and $\{gender, credit\}$, each with probability $1/3$. Γ_w is uniform over $\{\emptyset, \{check\text{-}failed\}\}$, each with probability $1/2$. In larger suspect sets the choice of cardinality range becomes substantive: sampling uniformly from the full powerset yields an expected intervention size of $|\mathbf{S}|/2$, which spreads attribution across large coalitions and may make individual feature contributions harder to isolate; we return to this design decision in the experimental section.

Causal attribution hinges on questions such as "if X had, contrary to fact, been fixed to $x' \neq x$ instead of x, what would the outcome have been?" For non-binary causes, this is ambiguous. To resolve this in a way that does not require committing to a single, contrastive causal event (Kawakami et al., 2024), we rely on a *distribution* over alternative values.

Definition 11 (Alternative Value Distribution) *Let* $\mathbf{X}$ *be a set of random variables with domain* $\mathrm{dom}\,(\mathbf{X})$ *and let* $\mathbf{X} = \mathbf{x} \in \mathrm{dom}\,(\mathbf{X})$ *be a factual event. An* alternative value distribution $\Delta(\mathbf{x})$ *is a probability measure on* $\mathrm{dom}\,(\mathbf{X})$ *such that* $\mathbf{x} \notin \mathrm{supp}(\Delta(\mathbf{x}))$.

Informally, $\Delta(\mathbf{x})$ *specifies what counts as a genuinely different value for a suspect variable. The support condition ensures no mass falls on the factual value* $\mathbf{x}$ *itself, so every sampled alternative is a real counterfactual departure rather than a noisy copy of what actually happened.*

When $\Delta(\mathbf{x})$ is a posterior conditioned on upstream factual values — the construction we use throughout — alternatives are weighted by their plausibility under the structural model with upstream context held fixed. This is the operational link to the descriptive-normality literature (Halpern and Hitchcock, 2015; Icard et al., 2017) mentioned in the introduction; PCI does not aim to resolve the counterexamples motivating that line of work, only to share its core intuition that causal attribution depends on what counts as a plausible counterfactual.

One natural alternative value distribution results from taking the one currently embodied in the model and excising it around the factual setting. Then, the semantics for "alternative value" means "a sufficiently different value, following the original distribution otherwise". In the discrete case, of course, the analogue is simply the probability mass function conditioned on the factual value not holding.

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practitioner to restrict the search to cause sets of a specific cardinality range: setting $J = K = 1$ confines attribution to individual variables, while $J = 2, K = 3$ focuses on small multi-variable interactions. Moreover, whereas exact Shapley computation requires summing over all $2^{|\mathbf{X}|}$ coalitions, PCI treats Γ as a generative model: approximation proceeds by sampling subsets from Γ and outcomes from $P_{\mathbf{U}}$, making Monte Carlo estimation a native feature of the framework rather than an external algorithmic layer. We note that the weighting alone does not distinguish PCI from SHAP at $J = 1$, $K = |\mathbf{X}|$: the subset weights are identical. The distinction lies in the integrand, and it matters for what is being evaluated and what results follow. SHAP averages marginal contributions using the observational conditional $\mathbb{E}[f(\mathbf{X}_S = \mathbf{x}_S, \mathbf{X}_{\bar{S}})]$, marginalising over the empirical distribution of non-suspect features; this means a variable that is merely correlated with a cause — but not itself causally active — can receive positive attribution. PCI instead evaluates counterfactual outcomes $P(Y \in A \mid \mathbf{u}, \mathrm{do}(\mathbf{C} = \mathbf{c}', \mathbf{T} = \mathbf{t}^\star))$ computed via the structural model, so attribution flows only along genuine causal paths. The witness mechanism adds a further distinction: by holding intermediate variables fixed, PCI can resolve overdetermination and undercutting scenarios where SHAP assigns equal or incorrect attribution. The choice of J and K remains a design decision; practitioners can set $J = K = 1$ for single-variable attribution or allow higher-order interactions as discussed above. A detailed comparison of PCI, SHAP, and Causal SHAP on concrete examples is given in Section 4.

Example 5 (ϵ-Excised Distribution) *Assume $\mathbf{X}$ is $\mathbb{R}^d$-valued with density p. Given a factual event $\mathbf{X} = \mathbf{x}$, define the ϵ-excised alternative value distribution $\Delta_\epsilon(\mathbf{x})$ as the probability measure on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ with density*

$$p_{\Delta_\epsilon}(\mathbf{x}' \mid \mathbf{x}) \propto \begin{cases} 0, & \mathbf{x}' \in B_\epsilon(\mathbf{x}), \\ p(\mathbf{x}'), & \textit{otherwise}, \end{cases}$$

where $B_\epsilon(\mathbf{x})$ denotes the ϵ-ball centered at $\mathbf{x}$. In particular, $\mathbf{x} \notin \operatorname{supp}(\Delta_\epsilon(\mathbf{x}))$.

Informally, the alternative value distribution Δ operationalises what "counterfactually different" means for each suspect variable: it specifies which alternative values are considered when asking whether the outcome would have changed under an alternative cause.

In the OBCB model both $gender$ and $credit$ are binary, so Δ is trivial. For Alice, the only alternative to $gender = 0$ (female) is $gender = 1$ (male), and the only alternative to $credit = 0$ (bad) is $credit = 1$ (good); for Bob, the only alternative to $gender = 1$ is $gender = 0$. Accordingly, $\Delta(gender = 0)$ is a point mass on $gender = 1$ and $\Delta(credit = 0)$ a point mass on $credit = 1$ — no distributional choice is required. The binary case is exactly where Δ is unambiguous; it is for continuous suspects such as income or loan amount that the ϵ-excised distribution becomes essential, forcing an explicit commitment to what counts as a meaningfully different value.[5]

We are now ready to state the main definitions of PCI. Because PCI is not constrained to particular variable types, we work in general measure-theoretic terms that support combinations of continuous and discrete variables. We use the interventional law (Definition 8) and marginalize over random samples of suspect sets, witness sets, and alternative values for the causal set, yielding a joint measure on factual and counterfactual outcomes.

5. Any method that asks "what would happen if $\mathbf{X}$ had been different" must operationalise what *different* means. Using the empirical distribution $p(\mathbf{x}')$ directly — as in kernel SHAP (Lundberg and Lee, 2017) — is not a neutral choice: because p places mass near the factual value $\mathbf{x}$, sampled alternatives may be practically indistinguishable from $\mathbf{x}$, conflating the necessity question ("what if $\mathbf{x}$ had genuinely differed?") with evaluating essentially the same setting. More generally, three natural choices exist for Δ. The observational marginal $p(\mathbf{x}')$ conflates the necessity question with the factual setting, as noted above. The fully conditional posterior $p(x'_k \mid \mathbf{x}_{-k})$ conditions on all other observed features — potentially including descendants of X_k — which imposes structural constraints that may rule out otherwise plausible counterfactuals, and still concentrates mass near x_k when features are correlated. The ϵ-excised distribution occupies the middle ground: it draws from the marginal informed by upstream variables only, excising the factual neighbourhood to guarantee genuine departure without downstream conditioning. The ϵ-excised distribution makes the commitment explicit and tunable: *different* means outside $B_\epsilon(\mathbf{x})$, with ϵ set by the practitioner to reflect the minimum change deemed meaningful. This parallels the Region of Practical Equivalence (ROPE) in Bayesian hypothesis testing (Kruschke, 2018): just as ROPE forces an explicit decision about what difference is large enough to matter, ϵ-excision forces an explicit decision about what counterfactual is sufficiently far from the factual to count as genuinely alternative. The practitioner makes this commitment regardless of method; the only question is whether it is made transparently or absorbed into a default. For practical guidance on choosing ϵ: when X is standardised to $\mathcal{N}(0,1)$, the choice acquires a direct probabilistic interpretation — the excised mass $\Phi(x^\star + \epsilon) - \Phi(x^\star - \epsilon)$ is the fraction of the distribution ruled out as insufficiently different from the factual value $x^\star$. At $x^\star = 0$, setting $\epsilon = 0.2$ excises approximately 15.9% of the distribution, $\epsilon = 0.5$ excises 38.3%, and $\epsilon = 0.8$ excises 57.6%, corresponding roughly to small, medium, and large effect thresholds in the sense of Cohen (1988). Alternatively, ϵ can reflect measurement precision (the smallest meaningful increment of X) or a decision-relevant threshold (the minimum change in X that would alter a downstream outcome). As with ROPE, a sensitivity analysis across a range of ϵ values is advisable when the appropriate scale is unclear. For concrete examples of how SHAP's implicit choice leads to problematic attributions, see Section 4.

Definition 12 (Joint Necessity and Sufficiency Measure) *Let $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$ be a probabilistic structural causal model and Y an outcome variable with domain $\mathrm{dom}\,(Y)$. Let p_s^Γ and p_w^Γ be the mass functions of variable selection distributions on $2^{\mathbf{S}}$ and $2^{\mathbf{W}}$, and let $\Delta(\mathbf{s}^\star)$ be an alternative value distribution on $\mathrm{dom}\,(\mathbf{S})$. Let $(\mathbf{S}, \mathbf{W}) = (\mathbf{s}^\star, \mathbf{w}^\star)$ be a factual event.*

For a variable of interest X_k, let $2_k^{\mathbf{S}} := \{\mathbf{C} \subseteq \mathbf{S} : X_k \in \mathbf{C}\}$, and for each $\mathbf{C} \subseteq \mathbf{S}$ let $\Delta_{\mathbf{C}}(\mathbf{s}^\star)$ denote the pushforward of $\Delta(\mathbf{s}^\star)$ under the restriction map $\mathbf{s}' \mapsto \mathbf{s}' \mid_{\mathbf{C}}$.[6] *For any measurable $A \subseteq \mathrm{dom}\,(Y)$ and exogenous realization $\mathbf{u}$, the* sufficiency *and* necessity *measures are defined as follows. The* sufficiency measure *$P_k^s(A \mid \mathbf{u})$ intervenes on each active suspect set $\mathbf{C}$ to its factual values $\mathbf{s}^\star \mid_{\mathbf{C}}$, holding active witnesses $\mathbf{T}$ fixed, and computes the Γ-weighted sum of $P(Y \in A \mid \mathbf{u}, \mathrm{do}(\cdot))$ under factual cause restoration. The* necessity measure *$P_k^n(A \mid \mathbf{u})$ instead integrates over alternative values $\mathbf{c}'$ for $\mathbf{C}$, again holding $\mathbf{T}$ fixed at factual values, and computes the Γ-weighted average of $P(Y \in A \mid \mathbf{u}, \mathrm{do}(\cdot))$ over alternative cause values.*

$$P_k^s(A \mid \mathbf{u}) = \sum_{\mathbf{C} \in 2_k^{\mathbf{S}}} \sum_{\substack{\mathbf{T} \in 2^{\mathbf{W}} \\ \mathbf{T} \cap \mathbf{C} = \emptyset}} P(Y \in A \mid \mathbf{u}, \mathrm{do}(\mathbf{C} = \mathbf{s}^\star \mid_{\mathbf{C}}, \mathbf{T} = \mathbf{w}^\star \mid_{\mathbf{T}}))\; p_s^\Gamma(\mathbf{C})\, p_w^\Gamma(\mathbf{T}), \tag{1}$$

$$P_k^n(A \mid \mathbf{u}) = \sum_{\mathbf{C} \in 2_k^{\mathbf{S}}} \sum_{\substack{\mathbf{T} \in 2^{\mathbf{W}} \\ \mathbf{T} \cap \mathbf{C} = \emptyset}} \int P\big(Y \in A \mid \mathbf{u}, \mathrm{do}\big(\mathbf{C} = \mathbf{c}', \mathbf{T} = \mathbf{w}^\star \mid_{\mathbf{T}}\big)\big)\, \Delta_{\mathbf{C}}(\mathbf{s}^\star)(d\mathbf{c}')\, p_s^\Gamma(\mathbf{C})\, p_w^\Gamma(\mathbf{T}). \tag{2}$$

The joint necessity-sufficiency measure then is given by

$$P_k^{s,n}(A \times B) \;:=\; \int P_k^s(A \mid \mathbf{u})\, P_k^n(B \mid \mathbf{u})\, P_{\mathbf{U}}(d\mathbf{u}). \tag{3}$$

$P_k^s(\cdot \mid \mathbf{u})$ and $P_k^n(\cdot \mid \mathbf{u})$ record, for a fixed noise realization $\mathbf{u}$, the outcome distributions under the sufficiency and necessity interventions respectively; marginalising over $P_{\mathbf{U}}$ recovers their marginal forms $P_k^s(A) = \int P_k^s(A \mid \mathbf{u})\, P_{\mathbf{U}}(d\mathbf{u})$ and analogously for P_k^n. The product decomposition in $P_k^{s,n}$ is legitimate because, conditional on $\mathbf{u}$, the sufficiency and necessity interventions operate in separate counterfactual regimes with no causal connection: $\mathrm{do}(\mathbf{C} = \mathbf{c}^\star)$ and $\mathrm{do}(\mathbf{C} = \mathbf{c}')$ act on the same noise but in distinct hypothetical worlds, so Y^s and Y^n are conditionally independent given $\mathbf{u}$, and $P(Y^s \in A, Y^n \in B \mid \mathbf{u}) = P_k^s(A \mid \mathbf{u})\, P_k^n(B \mid \mathbf{u})$.

Applied to the OBCB model with the choices established above — $\mathbf{S} = \{gender, credit\}$, $\mathbf{W} = \{check\text{-}failed\}$, Γ_s and Γ_w uniform, Δ point masses on alternative binary values, outcome $y^\star = loan = 0$ — Table 2 reports the expected causal impact for Alice and Bob under PNS, PCI without witnesses, and PCI with the *check-failed* witness active.

For Alice, PNS and PCI without witnesses both fail **D-A-rank** (p. 10): they rank *credit* above *gender* (0.180 vs. 0.045; 0.240 vs. 0.210). PCI with witnesses reverses this (0.263 vs. 0.199), satisfying **D-A-rank** as well as **D-A1** and **D-A2** (both values positive). Holding *check-failed* at its factual value blocks the credit-check bottleneck in noise realizations where Alice was checked, isolating gender's structural role. The *check-failed* witness is the decisive ingredient.

For Bob, **D-B2** and **D-B-rank** hold under all three methods: *credit* is ranked above *gender* throughout. **D-B1** requires strictly positive attribution for *gender*: PNS returns exactly 0.000 and

6. For an assignment $\mathbf{x}$ to a collection of variables $\mathbf{X}$ and any subset $\mathbf{Y} \subseteq \mathbf{X}$, the restriction $\mathbf{x} \mid_{\mathbf{Y}}$ denotes the sub-assignment of $\mathbf{x}$ to the variables in $\mathbf{Y}$.

Person	Feature	PNS	PCI (no witnesses)	PCI (with witnesses)
Alice	$gender$	0.045	0.210	**0.263**
Alice	$credit$	0.180	0.240	**0.199**
Bob	$gender$	0.000	0.038	**0.019**
Bob	$credit$	0.860	0.228	**0.119**

Table 2: Expected causal impact of $gender$ and $credit$ on $loan = 0$ for Alice (female, bad credit) and Bob (male, bad credit) in the stochastic OBCB model. PNS uses no suspect or witness structure; PCI without witnesses sets $\mathbf{W} = \emptyset$; PCI with witnesses sets $\mathbf{W} = \{check\text{-}failed\}$.

therefore fails, because $P(loan = 1 \mid gender = \text{female}, credit = \text{bad}) = 0$ makes the counterfactual necessity calculation zero, missing gender's indirect role in opening the credit evaluation. PCI with witnesses returns 0.019, correctly capturing this indirect contribution. The cross-individual desideratum **D-comp** holds under both PCI ($0.263 > 0.019$) and PNS ($0.045 > 0.000$). A full verification of all seven desiderata is given at the end of this section.

With the necessity-sufficiency measure fully defined, we are ready to characterize the causal impact of an event X_k. In particular, we will proceed by defining an arbitrary *causal impact function*, whose expectation over the necessity-sufficiency measure determines the causal impact. We first proceed with a formal definition of the causal impact measure, followed by several intuitive candidates for the causal impact function.

Definition 13 (Causal Impact Function and Its Expectation) *Let $y^\star \in \operatorname{dom}(Y)$ denote the factual outcome value, and let $Y^s, Y^n \sim P_k^{s,n}$. For any measurable function $ci : \operatorname{dom}(Y)^3 \to \mathbb{R}$, the* (expected) causal impact *of X_k is defined as*

$$\mathbb{E}[\, ci(Y^s, Y^n, y^\star)] \;:=\; \iint ci(y^s, y^n, y^\star)\, P_k^{s,n}(dy^s, dy^n). \tag{4}$$

Informally, ci is a user-defined scoring rule that translates the joint sufficiency–necessity measure into a single scalar. The choice is not arbitrary: the Absolute Difference example below is an instance of L1 distance between outcomes, the same metric underlying Average Treatment Effect, Conditional Treatment Effect, and SHAP. The binary indicator score recovers this as a special case when outcomes are binary — indicators are L1 on $\{0, 1\}$. Practitioners with reasons to prefer L2 or another distance measure may substitute it freely; the framework places no restriction on ci beyond measurability.

For the OBCB analysis we instantiate ci as the PNS binary score defined in the following example, with $y^\star = loan = 0$. This is the choice that produces the values in Table 2.

Example 6 (PNS for Binary Outcomes) *Assume the outcome Y is binary with $\operatorname{dom}(Y) = \{0, 1\}$, and let $y^\star = 1$ denote the factual outcome. Let Y^s and Y^n denote the sufficiency and necessity outcome variables. Define the causal impact score*

$$ci(y^s, y^n, y^\star) = \mathbb{I}\{y^n \neq y^\star\}\, \mathbb{I}\{y^s = y^\star\}.$$

Then the expected causal impact reduces to

$$\mathbb{E}[ci(Y^s, Y^n, 1)] = \mathbb{P}(Y^n = 0,\, Y^s = 1),$$

which conceptually coincides with Pearl's probability of necessity and sufficiency (PNS). The alignment is exact *when* $\mathbf{S} = \{X_k\}$*,* $\mathbf{W} = \emptyset$*, and* $\mathrm{dom}\,(X_k) = \{0, 1\}$ *where, factually,* $x_k^\star = 1$*.*

The causal impact function ci is a user-defined component of PCI , and different choices encode different causal questions. In the benchmarking experiments of Section **??**, we instantiate ci using only the necessity component, since Halpern's actual causality has no sufficiency counterpart — necessity-only attribution allows a direct parallel comparison. The full framework incorporating both Y^s and Y^n is instantiated in the PNS binary example above and in the Absolute Difference example below; the OBCB analysis of this section (Tables 2–3) provides a concrete validation, since the PNS binary ci jointly conditions on $Y^s = y^\star$ and $Y^n \neq y^\star$ and it is precisely this combination that produces the correct **D-A-rank** ordering. Further experiments using the full ci are presented in Section **??**.

Add cross-reference to sufficiency experiments once written.

Example 7 (Absolute Difference Impact Score) *We can develop an analogous measure for continuous outcomes where we increase attribution with distance from the factual value* $y^\star$ *in the necessity world and decrease impact with distance to the factual value in the sufficiency world.*

$$ci(y^s, y^n, y^\star) = |y^n - y^\star| - |y^s - y^\star|.$$

With the full machinery in place — structural model, variable selection, alternative value distribution, joint necessity-sufficiency measure, and causal impact function — PCI delivers the following properties.

(i) **Generalisation of PN/PS/PNS.** When $\mathbf{S} = \{X_k\}$, $\mathbf{W} = \emptyset$, and $\mathrm{dom}\,(X_k) = \{0, 1\}$, the PNS binary ci recovers Pearl's probability of necessity and sufficiency exactly; PN and PS are recovered by the corresponding one-sided ci choices.

(ii) **Connection to actual causality.** For suitable Γ and Δ, PCI recovers Halpern's actual causality judgements (Section **??**).

(iii) **Necessity–sufficiency decomposition.** P_k^s and P_k^n are defined and interpretable independently; practitioners may choose a ci that uses either alone or both, depending on whether the causal question concerns necessity, sufficiency, or both.

(iv) **Witness-mediated symmetry breaking.** Holding intermediate variables fixed via $\mathbf{W}$ disambiguates overdetermination and undercutting scenarios where necessity-only methods assign equal or zero attribution, as demonstrated above for Alice and Bob.

(v) **Composability.** Γ, Δ, and ci are probability distributions and a measurable function respectively; they slot naturally into Bayesian pipelines, accept posteriors as inputs, and admit Monte Carlo estimation without modification to the core formalism.

Table 3 verifies the seven desiderata introduced at the start of this section (p. 10) against the values in Table 2. PCIwith the *check-failed* witness satisfies all seven; PNS fails **D-A-rank** and **D-B1**. (The table uses the shorthand $|R(V \mid \text{person})|$ for $|R(V \rightsquigarrow \mathit{loan} \mid \text{person})|$.)

Desideratum	PCI (with witnesses)	PNS
D-A1: $\lvert R(\mathit{gender} \mid \text{Alice})\rvert > 0$	✓ 0.263	✓ 0.045
D-A2: $\lvert R(\mathit{credit} \mid \text{Alice})\rvert > 0$	✓ 0.199	✓ 0.180
D-A-rank: $\lvert R(\mathit{gender} \mid \text{Alice})\rvert > \lvert R(\mathit{credit} \mid \text{Alice})\rvert$	✓ $0.263 > 0.199$	× $0.045 < 0.180$
D-B1: $\lvert R(\mathit{gender} \mid \text{Bob})\rvert > 0$	✓ 0.019	× 0.000
D-B2: $\lvert R(\mathit{credit} \mid \text{Bob})\rvert > 0$	✓ 0.119	✓ 0.860
D-B-rank: $\lvert R(\mathit{credit} \mid \text{Bob})\rvert > \lvert R(\mathit{gender} \mid \text{Bob})\rvert$	✓ $0.119 > 0.019$	✓ $0.860 > 0.000$
D-comp: $\lvert R(\mathit{gender} \mid \text{Alice})\rvert > \lvert R(\mathit{gender} \mid \text{Bob})\rvert$	✓ $0.263 > 0.019$	✓ $0.045 > 0.000$

Table 3: Verification of the seven desiderata (p. 10) against the values in Table 2. PCIwith the $\mathit{check\text{-}failed}$ witness satisfies all seven; PNS fails **D-A-rank** and **D-B1**.

*PNS fails **D-A-rank** and **D-B1**. The D-A-rank failure reflects that without a $\mathit{check\text{-}failed}$ witness, necessity-based attribution cannot distinguish Alice's gender (which blocked the evaluation entirely) from her credit (which would only have mattered had she been evaluated). The D-B1 failure reflects that PNS assigns Bob's gender exactly zero: because $P(\mathit{loan} = 1 \mid \mathit{female}, \mathit{bad}) = 0$, the counterfactual necessity calculation returns zero, missing gender's indirect role in opening the credit evaluation. The $\mathit{check\text{-}failed}$ witness resolves both.*

4. Examples and comparison to SHAP and Causal SHAP

Shapley values have become one of the most widely used tools for explaining the predictions of complex machine learning models. Grounded in cooperative game theory, they provide a principled decomposition of a model's prediction into additive contributions from individual features. Heskes et al. (2020) propose Causal SHAP, which refines this by replacing the observational marginal used for features outside the active coalition with an interventional distribution: when a dropped feature is causally downstream of a coalition member, its distribution under the intervention differs from its marginal, and Causal SHAP corrects for this using Pearl's do-calculus. Formal definitions of both are given in Section 4.1.

Despite this repair, both methods share deeper limitations. First, both plain and Causal SHAP can assign non-zero responsibility to variables that play no causal role with respect to the target — attribution can run against the direction of the causal graph, a failure the interventional correction does not address. Second, both methods explain a model's predicted output averaged over the population, not the individual factual outcome of a specific person: when the goal is *causal responsibility attribution*, this produces further failures that are not always made explicit in the literature.

We trace these failures through two examples, comparing plain SHAP, Causal SHAP, and PCI against the desiderata of Section 2. We first return to the OBCB model: it permits exact analytic computation and the desiderata are already known. In OBCB the feature set contains only exogenous roots, so plain and Causal SHAP coincide — the comparison is short. We then turn to a continuous mediation example where downstream variables are observable model inputs; there the two methods diverge, and we examine what Causal SHAP fixes and what it does not.

4.1. SHAP and Causal SHAP

Definition 14 (SHAP) *A cooperative game is a pair (N, v), where $N = \{1, \ldots, n\}$ is the set of players and $v : 2^N \to \mathbb{R}$ is the* characteristic function *mapping each subset (coalition) $S \subseteq N$ to* *the value that coalition can produce. The Shapley value (Shapley, 1953) is the unique attribution satisfying four axioms:*

1. ***Efficiency**: $\sum_i \phi_i = v(N) - v(\emptyset)$*
2. ***Symmetry**: If $v(S \cup \{i\}) = v(S \cup \{j\})$ for all S, then $\phi_i = \phi_j$*
3. ***Dummy**: If $v(S \cup \{i\}) = v(S)$ for all S, then $\phi_i = 0$*
4. ***Linearity**: For two games v_1, v_2: $\phi_i(\alpha_1 v_1 + \alpha_2 v_2) = \alpha_1 \phi_i(v_1) + \alpha_2 \phi_i(v_2)$*

The unique solution is:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \big[v(S \cup \{i\}) - v(S)\big] \tag{5}$$

The weight $\frac{|S|!(|N|-|S|-1)!}{|N|!}$ equals the probability that, in a uniformly random ordering of all players, exactly the members of S precede i. Equivalently, ϕ_i is the average marginal contribution of i across all orderings. Lundberg and Lee (2017) apply Shapley values to explain model predictions by taking players to be input features, and the characteristic function to be the expected model output given a subset of features:

$$v(S) = \mathbb{E}_{X_{\bar{S}}}\big[f(X_S = x_S,\ X_{\bar{S}})\big] \tag{6}$$

where $\bar{S} = N \setminus S$ and missing features are marginalized over their marginal distribution.

Causal SHAP replaces the characteristic function with an interventional expectation (Heskes et al., 2020):

Definition 15 (Causal SHAP)

$$v_{causal}(S) = \mathbb{E}\big[f(X) \mid do(X_S = x_S)\big] = \int P(X_{\bar{S}} \mid do(X_S = x_S))\, f(X_{\bar{S}}, x_S)\, dX_{\bar{S}} \tag{7}$$

By the rules of do-calculus, intervening on X_S only affects the distribution of descendants of X_S. For non-descendants, $P(X_j \mid do(X_S = x_S)) = P(X_j)$. Writing $X_{\bar{S},\,\mathrm{nd}}$ for the features in $\bar{S}$ that are not descendants of any member of S, and $X_{\bar{S},\,\mathrm{d}}$ for those that are, equation (7) is equivalent to:

$$v_{causal}(S) = \mathbb{E}_{X_{\bar{S},\,\mathrm{nd}} \sim P(\cdot),\ X_{\bar{S},\,\mathrm{d}} \sim P(\cdot \mid do(X_S = x_S))}\big[f(X_{\bar{S}}, x_S)\big] \tag{8}$$

The Shapley formula (5) is otherwise unchanged. The method retains all four Shapley axioms.

4.2. OBCB: SHAP and Causal SHAP

We first apply SHAP to $g(x) = P(loan = 0 \mid x)$ for Alice and Bob from Example 3, attributing responsibility for the denial outcome.[7] Table 4 compares the resulting values against PNS and PCI.

7. We use $g = 1 - f$ where $f(g, c) = P(loan = 1 \mid g, c)$ takes values $f(\mathrm{F}, \mathrm{bad}) = 0$, $f(\mathrm{F}, \mathrm{good}) = 0.180$, $f(\mathrm{M}, \mathrm{bad}) = 0.045$, $f(\mathrm{M}, \mathrm{good}) = 0.900$. Because $g = 1 - f$, the characteristic functions satisfy $v_g(S) = 1 - v_f(S)$ and $\phi_i^g = -\phi_i^f$; we compute ϕ^f and flip signs. With $|N| = 2$ and uniform marginals $P(gender) = P(credit) = \frac{1}{2}$, equation (5) reduces to $\phi_i = \frac{1}{2}[v(\{i\}) - v(\emptyset)] + \frac{1}{2}[v(N) - v(N \setminus \{i\})]$. Population baseline: $v(\emptyset) = 0.281$. Alice: $v(\{gender = \mathrm{F}\}) = 0.090$, $v(\{credit = \mathrm{bad}\}) = 0.023$, $v(N) = 0.000$; hence $\phi^g_{gender} = 0.107$, $\phi^g_{credit} = 0.174$. Bob: $v(\{gender = \mathrm{M}\}) = 0.473$, $v(N) = 0.045$; hence $\phi^g_{gender} = -0.107$, $\phi^g_{credit} = 0.343$.

Person	Feature	PNS	SHAP	PCI (no witnesses)	PCI (with witnesses)
Alice	*gender*	0.045	0.107	0.210	**0.263**
Alice	*credit*	0.180	0.174	0.240	**0.199**
Bob	*gender*	0.000	−0.107	0.038	**0.019**
Bob	*credit*	0.860	0.343	0.228	**0.119**

Table 4: Responsibility attributions for Alice and Bob. SHAP values are signed (ϕ_i^g for $P(loan = 0 \mid x)$); desiderata compare magnitudes $|\cdot|$. PNS and PCI values from Table 2.

SHAP fails two desiderata. The primary failure is **D-A-rank**: credit outranks gender ($0.174 > 0.107$). This is an overdetermination effect: $credit$=bad correlates with the $check\text{-}failed$ mechanism that blocked Alice's credit evaluation, so SHAP's observational marginalization inflates credit attribution above gender's. PCI conditions on the factual $check\text{-}failed$ value directly, isolating gender's upstream causal role and recovering the correct ranking. PNS fails D-A-rank for the same reason; it also fails **D-B1** by returning exactly zero for Bob's gender (see Table 3).

The second SHAP failure is **D-comp**: $|\phi^g_{gender}| = 0.107$ for both Alice and Bob. But gender played a strictly larger causal role in Alice's rejection than in Bob's. For Alice, $gender$=female was sufficient on its own to produce rejection — it blocked the credit evaluation entirely, so her credit was never a factor. For Bob, $gender$=male was a facilitating condition: it triggered the credit check, but $credit$=bad was the actual driver of the denial. SHAP cannot distinguish these because it measures how far each feature value deviates from the population prediction mean — female and male are equidistant deviations in opposite directions, so they receive equal magnitude regardless of their individual causal weight. This is an instance of a deeper limitation: SHAP explains predicted outputs, not individual factual outcomes, which we examine further in Section 4.3.

Causal SHAP. In this example Causal SHAP coincides exactly with plain SHAP. The feature set $N = \{gender, credit\}$ consists entirely of exogenous roots: neither is causally downstream of the other, so $P(credit \mid do(gender = g)) = P(credit)$ and vice versa. The downstream variables ($check$, $check\text{-}failed$, $loan\text{-}if\text{-}checked$) are not observable as model inputs (while PCI can still consider the consequences of keeping them fixed as witnesses as the model runs unfold[8]); they are integrated out inside f. Because no dropped feature is a descendant of any coalition member, $v_{\text{causal}}(S) = v_{\text{plain}}(S)$ for every coalition, and the Causal SHAP values are identical to those in Table 4. In the signal example (Section 4.3), downstream variables are observable model inputs and the two methods diverge.

4.3. Signal with mediation

Example 8 (Signal with mediation) *We work with the following* ***generative model****: X sends a signal, M mediates it with some noise, and Y is the noisy final outcome:*

$$X \sim \mathcal{N}(0.5,\ 0.25) \tag{9}$$

$$M = X + \varepsilon_M, \quad \varepsilon_M \sim \mathcal{N}(0,\ 0.1) \tag{10}$$

$$Y = M + \varepsilon_Y, \quad \varepsilon_Y \sim \mathcal{N}(0,\ 0.1) \tag{11}$$

8. With the next example we will see that when downstream variables are observable model inputs, causal SHAP does not coincide with plain SHAP, and yet it still runs into trouble

The causal graph is $X \to M \to Y$, a simple mediation chain. All noise terms are independent of each other and of X. Since everything is jointly Gaussian, all conditional expectations are linear functions of the conditioning variables — no approximations are required.

Let $R(X \rightsquigarrow Y)$ denote the causal responsibility of X for Y. The ***qualitative causal responsibility desiderata*** *for this example are:*

$$
\begin{aligned}
&\textbf{\textit{DXY}} \quad R(X \rightsquigarrow Y) > 0\\
&\textbf{\textit{DMY}} \quad R(M \rightsquigarrow Y) > 0\\
&\textbf{\textit{DXM}} \quad R(X \rightsquigarrow M) > 0\\
&\textbf{\textit{DYX}} \quad R(Y \rightsquigarrow X) = 0\\
&\textbf{\textit{DYM}} \quad R(Y \rightsquigarrow M) = 0\\
&\textbf{\textit{DMX}} \quad R(M \rightsquigarrow X) = 0
\end{aligned}
$$

A somewhat more subtle desideratum is:

$$\textbf{\textit{DMXY}} \quad R(M \rightsquigarrow Y) > R(X \rightsquigarrow Y)$$

This captures the intuition that responsibility should be higher for variables that are closer to the outcome in the causal graph.

A further desideratum concerns the realized outcome rather than the model's prediction:

$$\textbf{\textit{D-ind}} \quad \sum_i \phi_i^B = B - \mathbb{E}[B]$$

The total attribution should account for the actual outcome deviation $B - \mathbb{E}[B]$ in the factual world, not merely the deviation $f_B(x) - \mathbb{E}[f_B(X)]$ of the prediction from its baseline. When residual noise drives B away from $f_B(x)$, an attribution method that explains only the prediction is silent about the gap.

The key ***distributional facts*** *we will use are:*

$$\mathbb{E}[X] = \mathbb{E}[M] = \mathbb{E}[Y] = 0.5, \tag{12}$$

$$\big(\textit{Var}(X),\, \textit{Var}(M),\, \textit{Var}(Y)\big) = (0.25,\, 0.35,\, 0.45), \tag{13}$$

$$\big(\textit{Cov}(X, M),\, \textit{Cov}(X, Y),\, \textit{Cov}(M, Y)\big) = (0.25,\, 0.25,\, 0.35). \tag{14}$$

For each target variable, the optimal predictor under squared loss is the conditional expectation, computable exactly in this Gaussian model:[9]

$$f_Y(X, M) = \mathbb{E}[Y \mid X, M] = M \quad (\{X, M\} \Rightarrow Y) \tag{15}$$
$$f_M(X, Y) = \mathbb{E}[M \mid X, Y] = 0.5X + 0.5Y \quad (\{X, Y\} \Rightarrow M) \tag{16}$$
$$f_X(M, Y) = \mathbb{E}[X \mid M, Y] = 0.5 + \frac{5}{7}(M - 0.5) \quad (\{M, Y\} \Rightarrow X) \tag{17}$$

4.4. SHAP values for the signal with mediation example

First consider a situation in which $X = 1, M = 1, Y = 1$. For each of those outcomes we will check our causal responsibility attribution desiderata against SHAP values. First, let's compute the SHAP values for X and M with respect to the prediction of Y.

Target Y, Features $\{X, M\}$. First, compute the coalition values:

$$v(\emptyset) = \mathbb{E}[M] = 0.5 \tag{18}$$
$$v(\{X\}) = \mathbb{E}[M \mid X = 1] = 1 \tag{19}$$
$$v(\{M\}) = \mathbb{E}[M \mid M = 1] = 1 \tag{20}$$
$$v(\{X, M\}) = f_Y(1, 1) = 1 \tag{21}$$

Next, apply equation (5) with $|N| = 2$:

$$\phi_X = \tfrac{1}{2}(1 - 0.5) + \tfrac{1}{2}(1 - 1) = 0.25 \qquad \phi_M = \tfrac{1}{2}(1 - 0.5) + \tfrac{1}{2}(1 - 1) = 0.25 \tag{22}$$

Target M, features $\{X, Y\}$. $v(\emptyset) = 0.5$; $v(\{X\}) = 1$; $v(\{Y\}) = 0.5\,\mathbb{E}[X \mid Y{=}1] + 0.5 = 0.889$ (using $\mathbb{E}[X \mid Y{=}1] = 0.5 + \frac{0.25}{0.45}(0.5) = 0.778$); $v(\{X, Y\}) = 1$.

$$\phi_X = \tfrac{1}{2}(1 - 0.5) + \tfrac{1}{2}(1 - 0.889) = 0.306, \tag{23}$$
$$\phi_Y = \tfrac{1}{2}(0.889 - 0.5) + \tfrac{1}{2}(1 - 1) = 0.194. \tag{24}$$

Target X, features $\{M, Y\}$. $v(\emptyset) = 0.5$; $v(\{M\}) = 0.5 + \frac{5}{7}(0.5) = 0.857$; $v(\{Y\}) = \mathbb{E}[X \mid Y{=}1] = 0.778$ (as above); $v(\{M, Y\}) = \mathbb{E}[X \mid M{=}1] = 0.857$ (since $X \perp Y \mid M$).

$$\phi_M = \tfrac{1}{2}(0.857 - 0.5) + \tfrac{1}{2}(0.857 - 0.778) = 0.219, \tag{25}$$
$$\phi_Y = \tfrac{1}{2}(0.778 - 0.5) + \tfrac{1}{2}(0.857 - 0.857) = 0.139. \tag{26}$$

Summing up, the SHAP values for the three targets are as follows:

9. **Distributional facts.** From the structural equations: $\mathrm{Var}(M) = \mathrm{Var}(X) + \mathrm{Var}(\varepsilon_M) = 0.25 + 0.1 = 0.35$; $\mathrm{Var}(Y) = \mathrm{Var}(M) + \mathrm{Var}(\varepsilon_Y) = 0.35 + 0.1 = 0.45$. $\mathrm{Cov}(X, M) = \mathrm{Cov}(X, X + \varepsilon_M) = \mathrm{Var}(X) = 0.25$; $\mathrm{Cov}(X, Y) = \mathrm{Cov}(X, M + \varepsilon_Y) = \mathrm{Cov}(X, M) = 0.25$; $\mathrm{Cov}(M, Y) = \mathrm{Cov}(M, M + \varepsilon_Y) = \mathrm{Var}(M) = 0.35$. **Conditional expectations.** $\mathbb{E}[Y \mid X, M] = M$ since $Y = M + \varepsilon_Y$ with $\varepsilon_Y \perp (X, M)$. For $\mathbb{E}[M \mid X, Y]$: the covariance matrix of (X, Y) is $\Sigma_{XY} = \left(\begin{smallmatrix} 0.25 & 0.25 \\ 0.25 & 0.45 \end{smallmatrix}\right)$ with $\det(\Sigma_{XY}) = 0.05$, so $\Sigma_{XY}^{-1} = 20\left(\begin{smallmatrix} 0.45 & -0.25 \\ -0.25 & 0.25 \end{smallmatrix}\right)$; the regression coefficients are $[\mathrm{Cov}(M, X),\ \mathrm{Cov}(M, Y)]\,\Sigma_{XY}^{-1} = [0.25,\ 0.35] \cdot 20\left(\begin{smallmatrix} 0.45 & -0.25 \\ -0.25 & 0.25 \end{smallmatrix}\right) = [0.5,\ 0.5]$, giving $\mathbb{E}[M \mid X, Y] = 0.5{+}0.5(X{-}0.5){+}0.5(Y{-}0.5) = 0.5X{+}0.5Y$. For $\mathbb{E}[X \mid M, Y]$: by d-separation, $X \perp Y \mid M$ in the chain $X \to M \to Y$, so $\mathbb{E}[X \mid M, Y] = \mathbb{E}[X \mid M] = 0.5 + \frac{\mathrm{Cov}(X, M)}{\mathrm{Var}(M)}(M - 0.5) = 0.5 + \frac{0.25}{0.35}(M - 0.5) = 0.5 + \frac{5}{7}(M - 0.5)$.

Target	Method	ϕ_X	ϕ_M	ϕ_Y
Y	Plain SHAP	0.25	0.25	—
M	Plain SHAP	0.306	—	0.194
X	Plain SHAP	—	0.219	0.139

Table 5: Plain SHAP attributions for instance $X = 1, M = 1, Y = 1$. For target X, plain SHAP assigns $\phi_Y = 0.139 > 0$, even though Y does not cause X. This arises because $v(\{Y\}) = 0.778 > 0.5$: observing $Y = 1$ is statistically informative about $X = 1$ through the chain $X \to M \to Y$. SHAP cannot distinguish "Y is informative about X" from "Y causes X" because the value function is grounded in observational conditionals, which are symmetric under reversal of causal direction.

4.5. Causal SHAP for the signal example

We now apply Causal SHAP (Definition 15) to the signal example and compare against plain SHAP.

Target Y, features $\{X, M\}$. $v_{\text{causal}}(\emptyset) = 0.5$; $v_{\text{causal}}(\{X\}) = \mathbb{E}[M \mid do(X{=}1)] = 1$ (M is downstream of X: $P(M \mid do(X{=}1)) = \mathcal{N}(1, 0.1)$); $v_{\text{causal}}(\{M\}) = 1$ ($do(M{=}1)$ sets $f_Y = M = 1$ directly, $P(X \mid do(M{=}1)) = P(X)$ since X has no causal parents); $v_{\text{causal}}(\{X, M\}) = 1$. All values match plain SHAP, so $\phi_X = \phi_M = 0.25$.

Target M, features $\{X, Y\}$. $v_{\text{causal}}(\emptyset) = 0.5$; $v_{\text{causal}}(\{X\}) = 1$ (Y is downstream of X, same as plain SHAP); $v_{\text{causal}}(\{Y\}) = \mathbb{E}_{X \sim P(X)}[0.5X + 0.5 \cdot 1] = 0.75$ (Y does not cause X, so $P(X \mid do(Y{=}1)) = P(X)$; differs from plain SHAP's 0.889); $v_{\text{causal}}(\{X, Y\}) = 1$.

$$\phi_X = \tfrac{1}{2}(1 - 0.5) + \tfrac{1}{2}(1 - 0.75) = 0.375, \tag{27}$$
$$\phi_Y = \tfrac{1}{2}(0.75 - 0.5) + \tfrac{1}{2}(1 - 1) = 0.125. \tag{28}$$

Target X, features $\{M, Y\}$. $v_{\text{causal}}(\emptyset) = 0.5$; $v_{\text{causal}}(\{M\}) = 0.857$ (Y is downstream of M but f_X does not depend on Y since $X \perp Y \mid M$, so $\mathbb{E}_{Y \sim P(Y \mid do(M=1))}[f_X(1, Y)] = f_X(1, \cdot) = 0.857$; same as plain SHAP); $v_{\text{causal}}(\{Y\}) = \mathbb{E}_{M \sim P(M)}[f_X(M, 1)] = 0.5$ (M is not a descendant of Y, so $P(M \mid do(Y{=}1)) = P(M)$ and $\mathbb{E}[f_X(M, 1)] = 0.5 + \frac{5}{7}(\mathbb{E}[M] - 0.5) = 0.5$; differs from plain SHAP's 0.778); $v_{\text{causal}}(\{M, Y\}) = 0.857$.

$$\phi_M = \tfrac{1}{2}(0.857 - 0.5) + \tfrac{1}{2}(0.857 - 0.5) = 0.357, \tag{29}$$
$$\phi_Y = \tfrac{1}{2}(0.5 - 0.5) + \tfrac{1}{2}(0.857 - 0.857) = 0. \tag{30}$$

4.6. Desiderata analysis for the signal example

Write ϕ_A^B for the attribution to feature A when predicting target B. Table 6 maps the desiderata from Section 4.3 to the values computed in Sections 4.4 and 4.5.

Positive results (DXY, DMY, DXM). Both methods correctly assign positive attribution to every direct or upstream cause of the target. Causal prediction models are built from observational conditionals, which do preserve correlation from cause to effect, so these desiderata pose no challenge to either method.

Causal SHAP fixes DYX, not DYM. For target X, the interventional value $v_{\text{causal}}(\{Y\}) = 0.5$ equals the baseline because M is not downstream of Y: $P(M \mid do(Y{=}1)) = P(M)$, so fixing $Y = 1$ provides no interventional leverage over f_X, and Causal SHAP correctly yields $\phi_Y^X = 0$. For

Desideratum	Condition	Plain SHAP	Causal SHAP
DXY	$\phi_X^Y > 0$	0.250 ✓	0.250 ✓
DMY	$\phi_M^Y > 0$	0.250 ✓	0.250 ✓
DXM	$\phi_X^M > 0$	0.306 ✓	0.375 ✓
DYX	$\phi_Y^X = 0$	0.139 ×	0.000 ✓
DYM	$\phi_Y^M = 0$	0.194 ×	0.125 ×
DMX	$\phi_M^X = 0$	0.219 ×	0.357 ×
DMXY	$\phi_M^Y > \phi_X^Y$	0.25=0.25 ×	0.25=0.25 ×
D-ind	$\sum_i \phi_i^X = X - \mathbb{E}[X]$	0.358 ×	0.357 ×

Table 6: Desiderata satisfied (✓) and violated (×) by plain SHAP and Causal SHAP on the signal with mediation example ($X \to M \to Y$), instance $X = M = Y = 1$.

target M, however, $\phi_Y^M = 0.125 > 0$ persists. The prediction model $f_M(X, Y) = 0.5X + 0.5Y$ contains Y as an input because Y is observationally predictive of M; setting $Y = 1$ raises the model output even under the interventional regime. Causal SHAP replaces the conditioning distribution but cannot remove a feature that the prediction model itself uses.

DMX worsens under Causal SHAP. Plain SHAP assigns $\phi_M^X = 0.219$; Causal SHAP assigns 0.357. Fixing DYX reduces $v_{\text{causal}}(\{Y\})$ from 0.778 to 0.5, driving ϕ_Y^X to zero, but reallocates the freed attribution to M. Since the structural noise that actually drives X is unobserved, all attribution concentrates on the only visible predictor, M, even though M does not cause X.

DMXY fails for both. Since $f_Y = M$, the singleton values $v(\{X\}) = \mathbb{E}[M \mid X{=}1] = 1$ and $v(\{M\}) = 1$ are identical, giving $\phi_M^Y = \phi_X^Y = 0.25$ for both methods. The Shapley formula cannot distinguish that X reaches Y only through M: when $X = 1$, the expected prediction is already 1 regardless of whether M is observed or not, so both features receive the same marginal contribution.

D-ind: predictions, not outcomes. For target X at the standard instance, $f_X(1, 1) = 0.857$ but $X = 1$. SHAP's local-accuracy axiom guarantees $\sum_i \phi_i = f_X(x) - f_0$, which evaluates to 0.357 at this instance — short of the actual deviation $X - \mathbb{E}[X] = 0.5$ by 0.143. Causal SHAP changes the conditioning distribution but inherits local accuracy, so it explains the same 0.357. The gap is the contribution of ε_X — the noise term that drove X above its prediction — and is invisible to both methods by construction. (For targets Y and M at this instance, f happens to equal the realized outcome, so D-ind is satisfied trivially; the failure is only visible when $f(x) \neq B$.)

The point is sharper at the instance $X = 0.5$, $M = 0.5$, $Y = 1$: features sit at their means, so $f_Y(0.5, 0.5) = 0.5 = \mathbb{E}[Y]$ and both methods assign $\phi_X = \phi_M = 0$. Yet $Y = 1$ deviates from baseline by 0.5, driven entirely by ε_Y. For *model* explanation this is correct — the prediction at baseline is the baseline. For *causal responsibility* attribution the implicit answer "no feature is responsible" is itself a substantive claim about ε_Y, one that SHAP and Causal SHAP are not equipped to articulate.

Overall, Causal SHAP strictly improves on plain SHAP for one desideratum (DYX) by replacing observational conditioning with interventional distributions. The remaining failures trace to three limitations neither method addresses: prediction models that include non-causal features (DYM, DMX), the inability of Shapley averaging to distinguish direct from indirect causal paths (DMXY), and the local-accuracy axiom that ties attribution to the prediction $f(x)$ rather than to the realized outcome (D-ind).

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